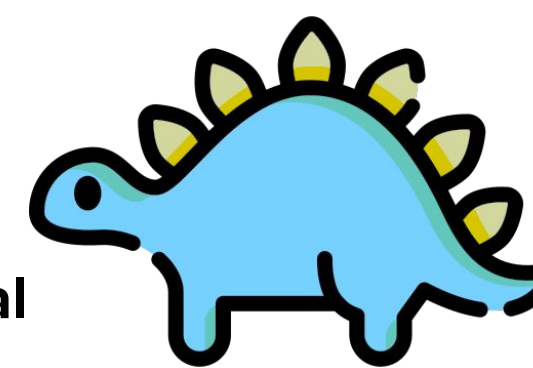
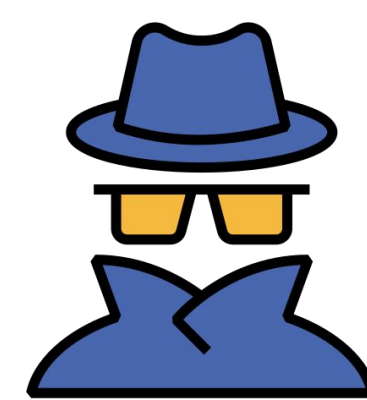
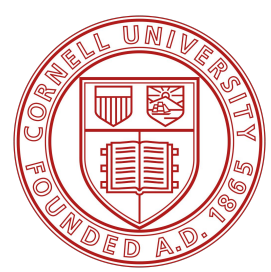




Motivation

Goal: Hide message in an image imperceptibly via Steganography using Deep Learning

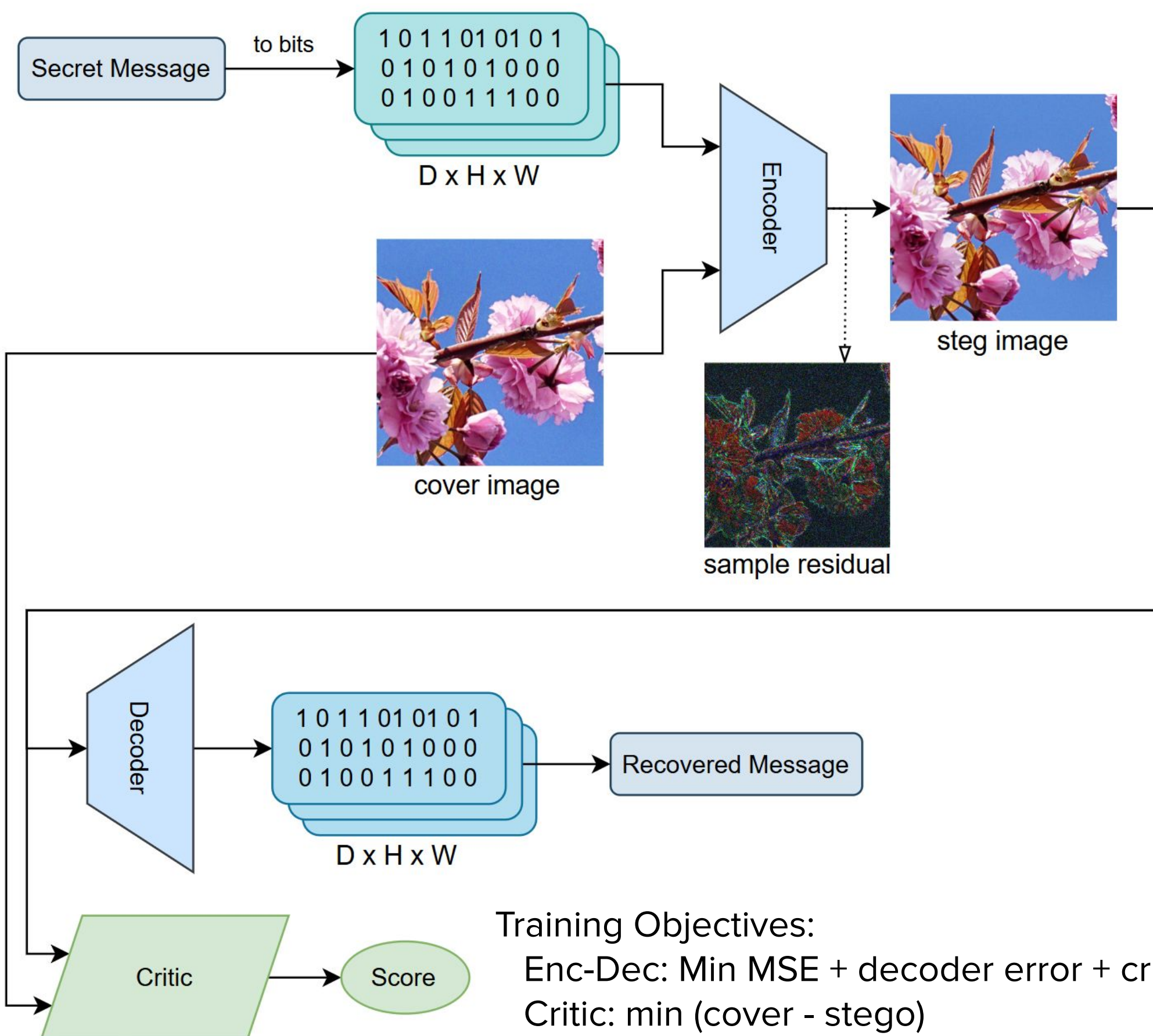


Traditional techniques are limited hand-crafted algorithms or achieve low message payloads (~0.4 bits-per-pixel).^[1]

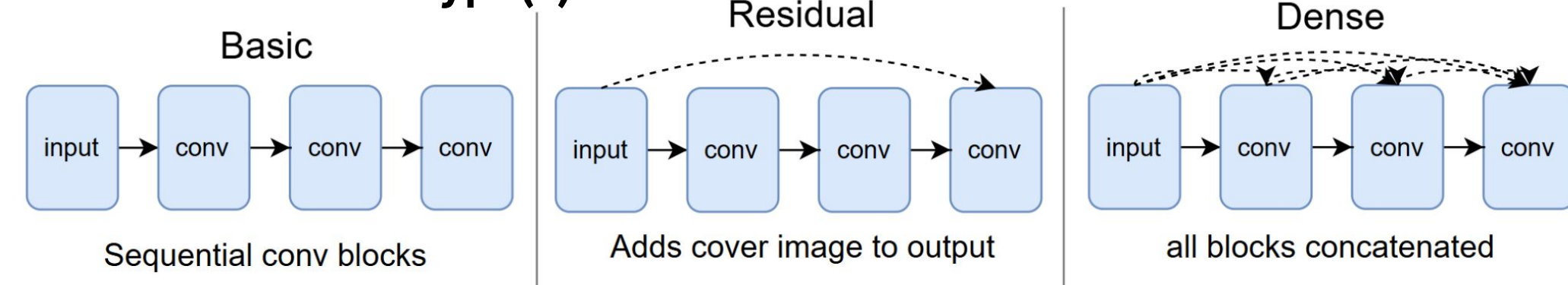
Why Steganography? (1) no need for private key sharing (2) private key makes presence of message obvious

Methodology

Model Architecture Natural Image Datasets: Div2K, COCO

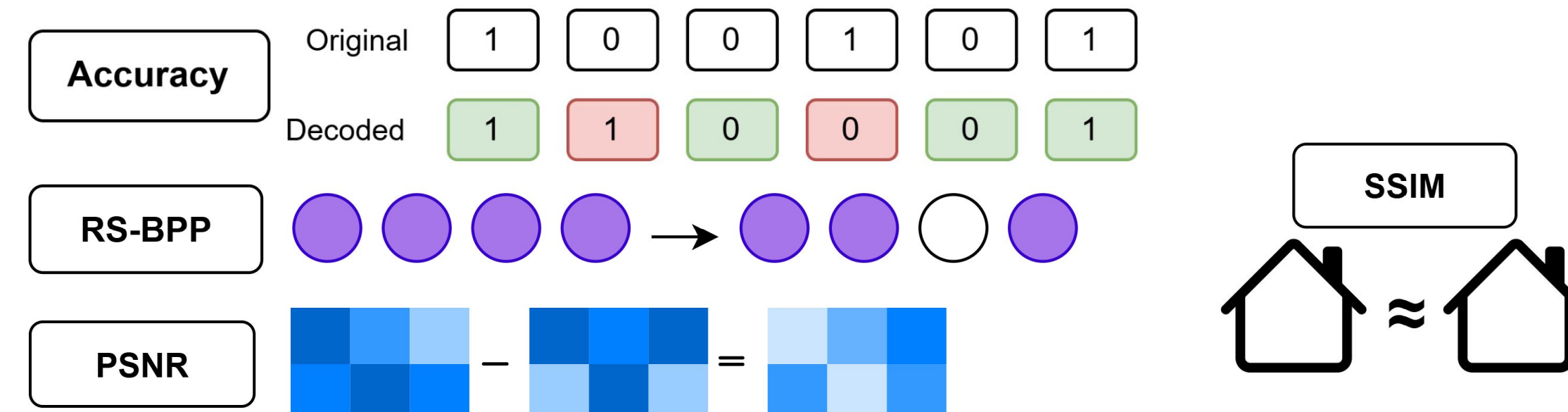


Encoder/Decoder Type(s)



Results

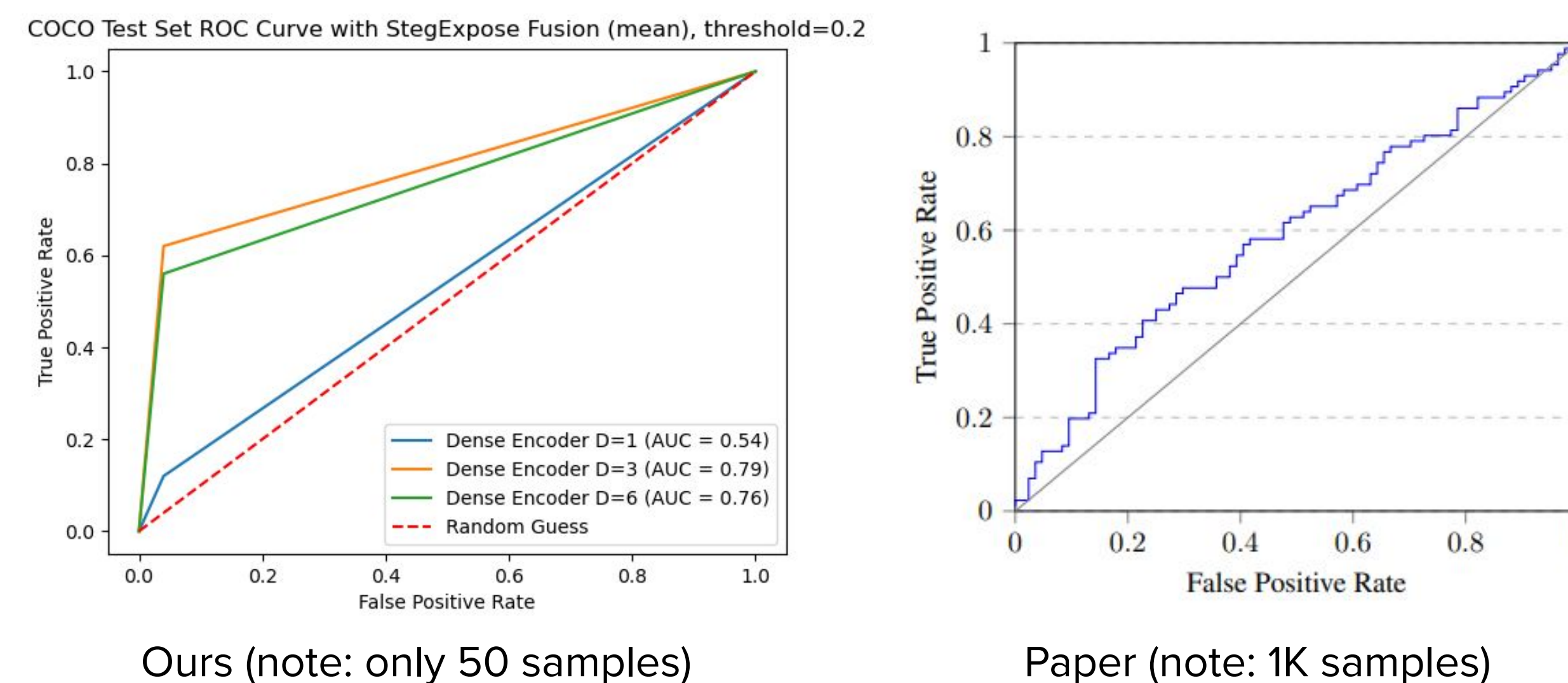
Metrics (RS-BPP: new metric created by authors)



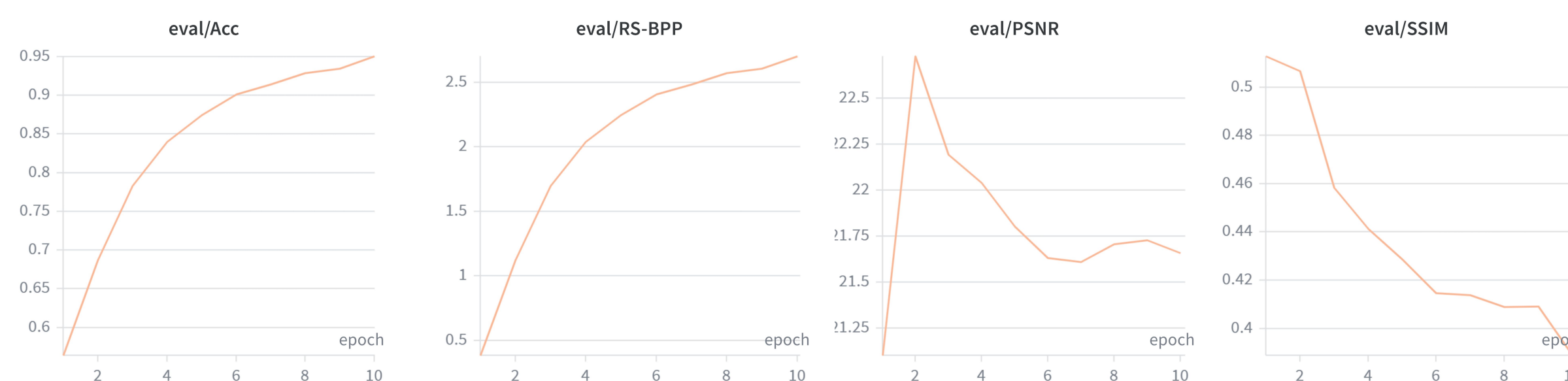
Experiment Results (Ran Depths (D) 1, 3, & 6)

Depth: 1	Accuracy				RS-BPP			
	Basic	Resid.	Dense	Attn	Basic	Resid.	Dense	Attn
Our results	0.96	0.95	0.96	1.00	0.94	0.89	0.92	0.99
Paper	0.95	0.99	1.00	-	0.91	0.99	0.99	-
	PSNR				SSIM			
	Basic	Resid.	Dense	Attn	Basic	Resid.	Dense	Attn
Our results	24.69	31.67	28.86	33.63	0.70	0.90	0.83	0.94
Paper	24.52	41.68	41.6	-	0.7	0.96	0.95	-

Evaluation With StegExpose Tool ^[2]

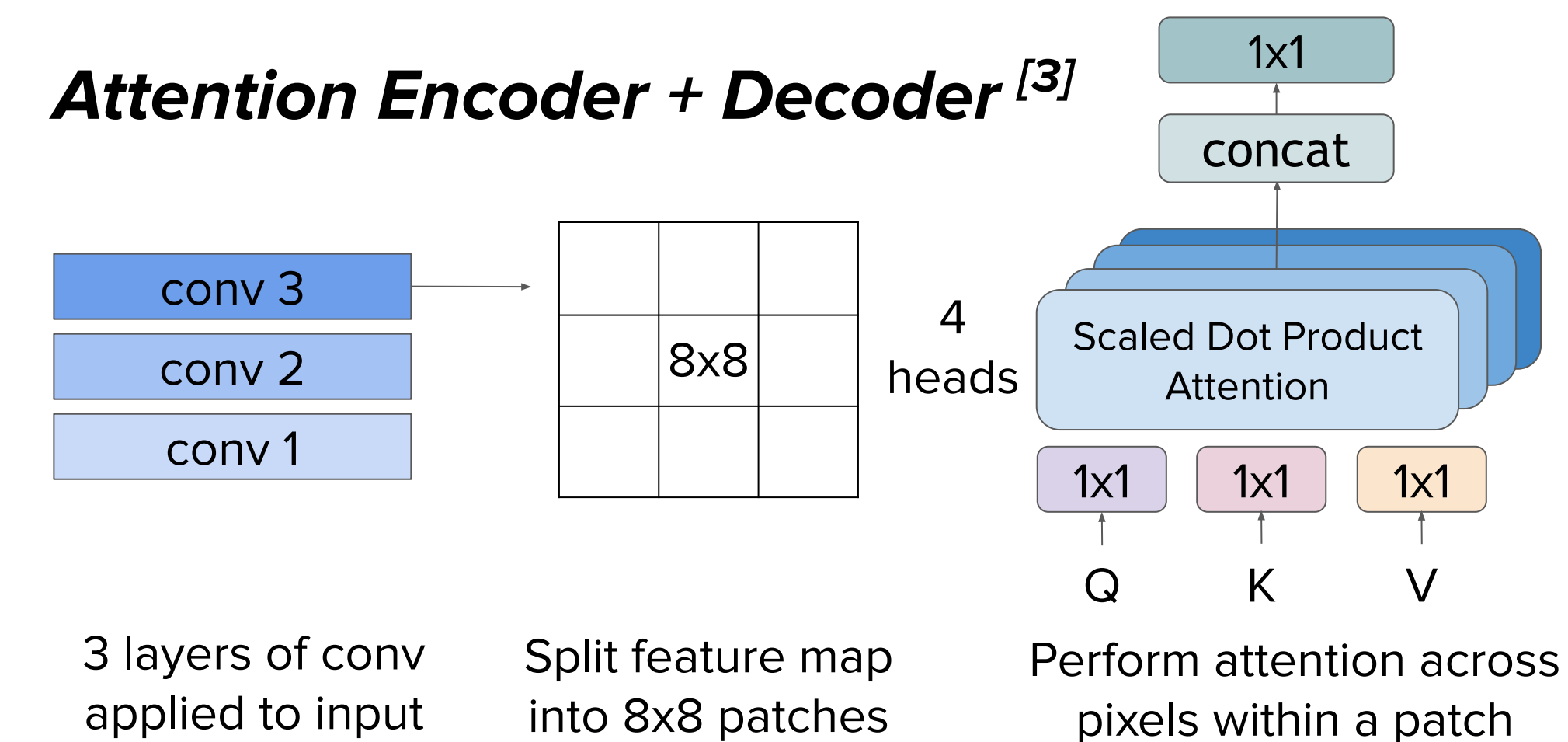


Metric Trade-offs Acc ↑, RS-BPP ↑, PSNR ↓, SSIM ↓ (>= pareto frontier)



Further Exploration

Attention Encoder + Decoder ^[3]

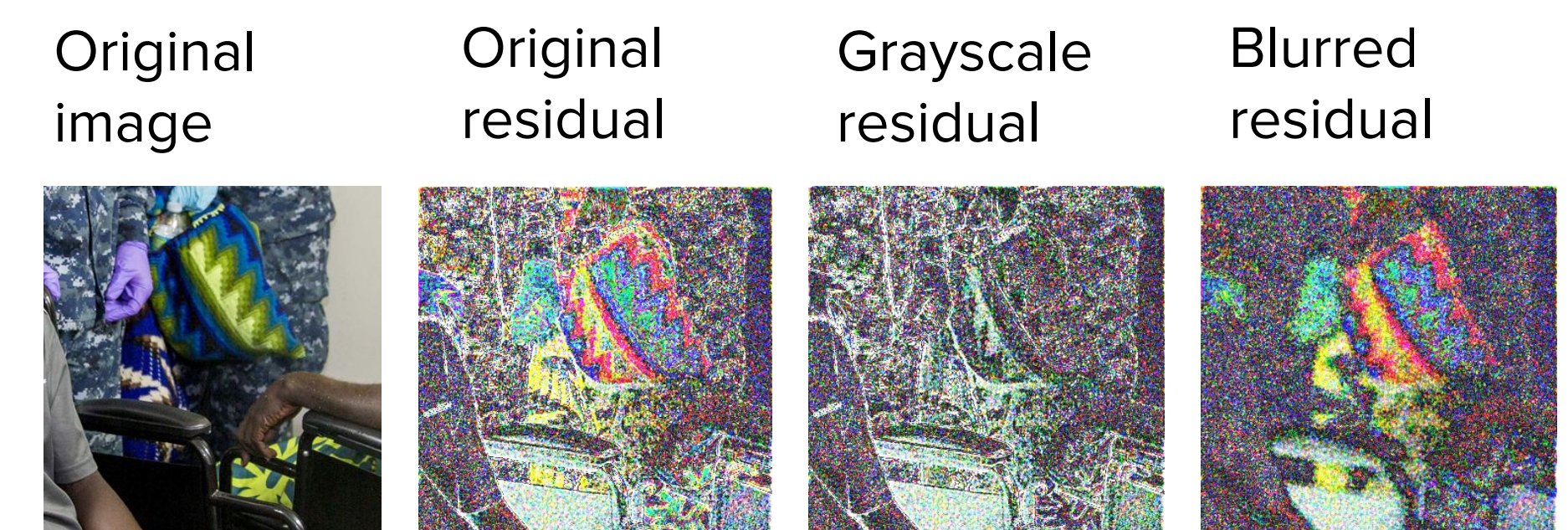


Scaling Loss Terms

Ld + **Ls** + Lr vs Ld + **100 * Ls** + Lr

Basic, D=3	Accuracy	RS-BPP	PSNR	SSIM
Regular Loss	0.86	1.76	22.45	0.53
Scaled Ls by 100	0.72	1.32	30.17	0.88

Evaluation on Grayscale, Blurred Images



Discussion/Future Work

- Achieved similar trends as paper
- Scaling up Attention Encoder + Decoder outperformed our results
- Model struggles to encode more bits without detection

Future Work

- Modify Critic loss
- Apply activation (tanh) to Dense and Residual
- Scale MSE loss (Ls) by constant for all experiments